

Zoetis Inc.

ANNOUNCER: Ladies and gentlemen, the program is about to begin. Reminder that you can submit questions at any time via the ask questions tab on the webcast page. At this time, it is my pleasure to turn the program over to your host, Michael [? Risken. ?]

SPEAKER 2: Great. Thanks for joining us. I'm Mike [? Risken, ?] analyst [INAUDIBLE] covering life science tools and diagnostics, and also covering the animal health space. Thanks for joining us. For our next session, we're pleased to host Zoetis, and we're joined by Wettey Joseph. CFO, Wendy, thanks for joining us.

WETTENY JOSEPH: Thanks for having me, Mike. I appreciate being on here with you.

SPEAKER 2: Great. Of course. As the operator said, feel free to submit questions via the portal. You can also send us [INAUDIBLE] chats or email and we'll work them into the call. Format will be fireside chat as usual. And maybe just kick things off, Wettey, your [INAUDIBLE] results recently, can you give us a brief recap of some of the key points, look back at 2022, look forward into 2023. What should we be thinking about?

WETTENY JOSEPH: Yeah. Look, reported strong results for 2022, finished the year strong with fourth quarter, operational revenue growth at 9% and adjusted net income operationally up 27%. That puts the year at 8% operational growth in revenue. 11% at adjusted net income.

Of course, our diverse portfolio, our global scale, and our innovation-driven products and franchises really allow us to grow faster than the market, and position us to be able to put 2023 into context, and be able to grow between 6% and 8% at top line. 7% to 9% operationally, at adjusted net income for 2023.

But clearly, the results for 2022 were at the high end of the revised guidance that we issued in November, and we're very pleased to deliver it and walk into 2023 with momentum across our products and so on.

SPEAKER 2: Great. Maybe I'll follow up on a couple of points there. First, you talked about the revised guidance for 2022. You guys updated the guide in 3Q to account for some of the supply chain disruptions, things like that.

And like you said, came at the high end of the revised guide. Can you give us an update on some of those supply chain issues? Are they fully resolved? Where are we as we're heading into 2023 on those factors?

WETTENY JOSEPH: Yeah, if you look across the categories that impacted from a supply constraints perspective, they were largely around parasiticides, both our TRIO product, and revolution, revolution plus. And really didn't have any particular supply constraints around derm, but we were constrained throughout the year in terms of our OA pain franchise, and Librela in particular, across Europe.

And as we exited 2022, we actually with confidence, that we have largely resolved all these challenges. In fact, we are in position to be able to run promotions for TRIO at the end of the year in the fourth quarter, having gotten supply back on track there, as well as revolution plus both for the US and China market, that are big markets.

So as we enter it into 2023, we're looking deliver to demand across our big areas. If you look at the largest growth drivers for us in 2023, you start with parasiticides. And, of course, TRIO is a big feature there, but we have a very broad portfolio that we're able to capitalize on demand across the globe on. As

well as dermatology. And not only are we able to continue to drive growth in dermatology, given \$6 million in the US and similar numbers outside the US that are untreated or undertreated, if you will.

We have the ability to also capitalize on demand that we see for Cytopoint, which historically, given some of the inputs are the same as used in COVID. We've not been able to fully capitalize on the demand there. And coming into 2023, all of that is behind us as well. So we feel great. And then across the OA pain franchise, of course. So we feel great coming into 2023 having solved-- largely solved some of these supply constraints.

I think you may see some intermittent issues around some of the livestock vaccines, but those are not going to move the needle for us in terms of coming into 2023 and these are all factored into the guidance that we gave just a couple of weeks ago.

SPEAKER 2: OK. That's really helpful. Maybe switching then to that guidance 6% to 8%, can you talk us about what you see happening to hit 8%? What goes wrong to be at 6%? What are the key puts and takes between the top and the bottom end of the range?

WETTENY JOSEPH: Look. Coming into 2023, of course, taking into account some macro uncertainty when you look at inflation and you look at consumer confidence and consumer spending. We factored those into our guidance, but we are in a very resilient industry, and we have innovative-driven products that help us grow. And you've seen that demonstrated through various market conditions over the last couple of decades for the industry, and Zoetis continues to grow above market.

So we factored those into our thinking for 2023, keeping an eye on the labor constraints across vet's practices as well. We factor in a bit more robust price increase in 2023. So in '22, we had 3% net price increase. And if you take Draxxin out, which had continued generic pressure, it was about 4%. And in 2023, we anticipate somewhere between 4 and 5% price, if you will, on a net basis across the business. So we factored those into our thinking. We are expecting competition for TRIO in the first half of the year, which I'm sure you and I will talk about a little bit more here in this segment. And as we continue to be confident that we will see an approval for Librela in the first half of 2023, given that the product is not yet approved, and we would not be launching until late in 2023, we did not include Librela in our guidance for 2023. That's our normal practice.

I do want to make sure that I underscore and reiterate a few points that we gave around phasing on the year because I want to make sure those are very clear in everyone's minds. We're very confident in delivering the guidance range that we gave. Again, 6% to 8% top line, 7% to 9% adjusted net income. But we do see a number of factors that are impacting the first quarter as we said in our prepared commentary a couple of weeks ago.

The Q1 growth rate will be below the low end of our range. So the low end of our range is 6%. We can probably see a Q1 growth rate that's probably around 4%, give or take, operationally. And in addition, as we said on the earnings call, FX really started to move last year in the second half. So the first half of the year, though FX, we expect to be roughly neutral on the full year as we go through 2023 based on where rates are hovering right now.

We do see headwinds in FX in the first half of the year, and the first quarter of this is about three points. So if you factor-- call it a 4% operational growth rate on Q1, you've got FX headwind of about 3% that puts you closer to a reported growth rate that's closer to 1%, if you will. But I do want to make sure that I give some additional color as well.

Again, very consistent with what we've factored into our guidance. So this is not a change, but I do want to make sure we reiterate that given the recovery timing of supply, which we were just talking about some of the supply challenges in 2022, the recovery of those at different points during the year in particular, in the fourth quarter, coupled with really, anticipation of price increases in 2023 caused distributors in particular, to buy up more in Q4, which will play itself out in the first quarter.

That's really particularly, with TRIO, that's a factor that impacts the US more so. The field force changes that we made last year starting in the second quarter, impacted diagnostics if you will, starting the second quarter. So the first quarter is a tougher comp with respect to that. And one more point is if you look at Draxxin, Draxxin is-- we're going to be lapping the second full year of Draxxin during our competition after the first quarter.

So that's another element here that impacts largely, the US. So as I think about the first quarter, I can see very strong growth coming from our international segment. But the US is likely to be flat to slightly down on the first quarter. We do expect the US to see strong growth for the year 2023, but this is the phasing that we factored into our guidance that we already gave and the comments that we gave two weeks ago. So this is in net difference.

A couple of things that I would say in terms of how we're seeing the end market demand. Right now, if you look at sales out of distribution, as well as sales into our direct channels with customers, we're seeing really positive trends there. We're seeing volume growth at those ends that are healthy volume growth in addition to obviously, price considerations that we have here as well. Although it's still early in the year, when we look at vet visits in January, we're actually starting to see an improvement in the trend up in the vet visits.

Again, it's still early in the year, a month in, if you will. We don't have data for February yet, but those are encouraging signs in terms of what we're seeing from an end market demand perspective, which underscores our confidence in the guidance that we gave for the full year. But I just wanted to make sure that the comments we already made around Q1 were well received, and that's why I've emphasized them here.

SPEAKER 2: Yeah. I appreciate that Wendy. There's a lot to unpack there. I was looking through my model as you were talking and I was like, most of what you called out, we have. So we're not egregiously off from one Q but yeah, agreed. That's what we called out on the call so-- I appreciate you emphasizing that.

I want to ask a little bit again, on the vet visit side of things. You talked about some stabilization, some recovery. And I think you can say in terms of what you're assuming in your guide if last year was like a 2% to 3% decline for vet visits. Are you assuming roughly flat this year for vet visits for the year in the US or?

WETTENY JOSEPH: Look. I think coming into the year, we're assuming a flat 2, 1%. Maybe more of that coming in the back half in terms of some positive trends. Maybe the front half would have been flat to slightly down, give or take. Those are some of the areas that we were looking at coming into this year. I do want to put into context of vet visits as we've done a number of times. Mike, I think if you look at vet visits in 2022, and you saw visits down anywhere from 3% to 4% depending on the quarter that you're looking at, those were against peak numbers in 2021.

But even as you look at Q4 vet visit numbers in the absolute, they were still about 2% above where they were before the pandemic. And really, we're talking about a comp against peaks in 2021. So our

expectation was that as we exited 2022, somewhere in the first half of 2023, we would see a bit of that stabilizing versus prepandemic and versus where they started to come down last year.

And we would expect to see that starts to turn to some positive visit somewhere in the middle to the back half of the year. Again, January is positive. Still early, so I don't want to move too far with that, but it is encouraging what we're seeing so far.

SPEAKER 2: Got it. That's helpful. And then the other point of the guide on price, 4% to 5%. You called out that excluding tracks, and you would have had 4% last year as well. Can you talk about that 4% to 5% split between companion and livestock? US, OUS? New products versus old products? How broad is it? And was it all taken on January 1, so it's already all in place, or how is that rolled out?

WETTENY JOSEPH: Yeah. Look, I think there isn't anything that I would point to in 2023 that's markedly different than what we typically do around pricing. I think you can expect the innovative products where we're continuing to see really strong demand to have a bit more price than some of the established products that have been in the market. It doesn't mean that we're not taking price across established products, it's just relatively speaking, a bit more on some of the newer, more innovative products than the others.

I think you can expect about the same level of price take across international versus the US. We don't put a blanket number. We do look at each market and each product when we come up with what price we take. And so it does vary quite a bit. But that's common practice that we take into consideration. And generally speaking, our price increases go in at the beginning of the year. There are times when we've taken mid-year price increases like last year in some of our international markets, we did that.

But generally speaking, when we talk about price, it tends to be early in the year. Now, the impact on price can sometimes be a little bit delayed depending on prepriced buy up that happens, which we talked about already. With respect to TRIO, that was one of the contributing factors, which we are in the fourth quarter in addition to our promotions and recovery of supply. So that tends to sometimes put a little bit of a push in terms of when the timing of that takes up but generally, we take prices up early in the year.

SPEAKER 2: Got it. And you touched on tracks and lapping the two year generic comp dynamic at the end of one Q. You talked about previously about first year, you help serve on tracks, and better than you anticipated. Second year was a little bit worse, and that you're roughly to where you expect it to be at the end of two years. Any update on what to think about for year three, year 4? You think it'll stabilize from here or how are you looking at it?

WETTENY JOSEPH: I think-- and you're right. Right? In 2021, we saw about 16% impact largely on price. We've actually been able to maintain most of our volume. We've seen a little bit of expansion of the macrolide class, which is a premium class, which has made up for some of it but obviously, the price take-- fewer price give in this case, has resulted in about 16% headwind in 21. In 22, the number was a bit more than the 20% that we said we would see each year.

So offsetting that favorability if we call it that, right? In terms of the first year. So probably about 25% headwind in the second year. We still have another quarter of that through Q1, to fully lap the second year.

Look, I won't signal here that we're not going to see some headwind. I just don't think it'll be to the magnitude that we've seen in these last two years. And you are at a level, in terms of the sales here, that it won't have as big of an impact on the overall picture given where it's come down from its peak level. So

I think we'll see some, but not the extent that we've seen in the past, which means it'll be more muted in terms of the impact for the overall Zoetis results.

SPEAKER 2: Great. Any pushback you noticed on price increases across the board, whether it's from your distributors or from your larger corporate clinics? How is that being received by the market? Two consecutive years of pretty solid price increases, what's the feedback on that?

WETTENY JOSEPH: Well, this is an area that we'll continue to monitor, but I would say, we're not seeing any pushback in this regard. We have innovative products that have been in the market for quite some time. You take [INAUDIBLE] for example, almost 10 years. Cytopoint, 5 years. These are products that we continue to take price on, and we continue to see volume growth.

Just remind you, in 2022, we had net 3% price. We had 5% volume growth in 2022, driven by these products that are just referenced with key term growing 17% operationally in 2022. And I would say a good high single-digit number of that is coming from price.

So I think we continue to see being underscored, what research tells us. If you look at the industry, it's very resilient. The premium on pet health is something that we have tested in terms of some of these times and time again. The HABRI study for Human Animal Bond Research Institute shows that 86% of pet owners are going to quote, unquote "spend whatever it takes" if their pet needed extensive veterinary care.

More recently, Zoetis' sponsored market study that-- in a hypothetical situation, if pet owners had a 20% reduction in their budget, what would that mean in terms of pet spend? And the short answer is, they would not change their pet spending. So this continues to be underscored. I would say, it's a resilient industry, and we continue to see that play out, and we continue to innovate.

And quite frankly, we talk a lot about vet visits, and they're important because we continue to innovate in products that are injectables and MABs, et cetera, that will go through the vet channel. So those are important areas for us. But in the context of what's driven industry growth year on year, visits have been about a 1% contribution to the growth rate of about 5% to 6% for the industry across 20 years.

Now, Zoetis is going above that, but that's really putting into context. And the reason that's important in the context of your question here, in terms of any kind of pushback around price increases down to the consumer, is that we continue to see spend per visit in revenue for the vet practices be up.

They're up about 25% from where they were in 2018, and you continue to see high single-digit to low double-digit growth in that. In the fourth quarter, revenue was up 6% for vets. And per visit was up 10% even as you saw visits down four. So I do think those continue to underscore the ability.

And by the way, vets are taking price on services at a level that's higher than products, and continue to pass price. And it's still difficult to get an appointment into the vet clinic. So I do think these are factors that we don't take for granted. We'll continue to monitor and watch those, but the resiliency has continued to prove itself out.

SPEAKER 2: Got it. That's really helpful. I want to pivot a little bit and move into some product-specific questions. We're getting a lot in the chat. So first, let's touch on Librela. I think it's in terms of underlying demand for Librela in Europe because it seems like you saw a quarter over quarter decline in Europe in the fourth quarter.

And that was a little bit unexpected given how strong its ramp, and how it's early in the lifecycle. So anything to touch on [INAUDIBLE] and thoughts for Librela in Europe specifically, this coming year.

WETTENY JOSEPH: Yeah. We continue to see really strong demand for Librela across our European markets, even before we talk about other markets across international, that we are launching into throughout 2023. So just to double click a little bit in terms of what we saw with Librela, if you take a look at the second half of 2022, and what our sales were for Librela versus the second half of 2021, there were nearly 100% growth in that.

So what we saw in the fourth quarter was that as we lifted the allocations that we were on for most of 2022, particularly in certain practices, they did raise up, in terms of their inventory levels in the third quarter that benefited Q3 a bit and took away from some of what we saw in Q4.

As we go into 2023, we expect the first quarter to be the highest sales figures for Librela in the first quarter of 23 to date. And that's just off of the existing European markets. We won't see contributions from the new market launches until the second quarter, which effectively begun a couple of days ago.

So our international business has a one-month lag in terms of close. And so, we're now into the second quarter, and this is the quarter where we'll start to see the launches in other markets. We're poised to launch in Brazil, Canada, Japan, and Australia through-- not all in Q2, but starting Q2 and into Q3.

And so those will be additive to the picture, but we're very confident what we're seeing in terms of patients share gains, volume. One other thing I would say is, if you take FX, for example, where we started the year to where we ended the year, keep in mind the second half of the year was when FX really moved on us. If FX were the same from the start of the year throughout the year, the number on Librela would have been an additional 17 to \$18 million.

So FX did play a role here. It doesn't factor into the operational picture because it's a launch product that's continuing to ramp. But I just think that's one other element to keep in mind here. But we're very confident in terms of what we're seeing. In terms of growth, Librela is already the number one paying product in Europe. Blockbuster status within the first full year, et cetera.

And now that we are off allocation and are able to supply to demand, we can also increase our start, if you will, DTC and other awareness campaigns that will drive patients into the clinic here on this product, which we were not in position to do for most of last year. Certainly, the first half from last year into starting the second half of last year as well.

SPEAKER 2: OK. And then Librela in the US, you still talk about first half approval. I think last time you spoke, you indicated that you've had the site visit and you're just waiting for the final stamp of approval. Can you talk us through steps after that? Let's say you get approval. I'm not going to pick [AUDIO OUT]. Let's see if we get approval on date X, what happens on X plus 1, X plus 2, X plus 90? Just walk us through the rest of the year from that point.

WETTENY JOSEPH: Yeah. Look, our plan for Librela hasn't changed, and I think the talk track here has been the same for the last three or four quarters. The only change that we made was to update that we have in fact, had those inspections at those sites outside the US. Those were successful inspections. And we are confident that we will see an approval for this product in the first half of 2023. In terms of our plans for launch, again, those that haven't changed. We will-- once we get the approval, that sets what the label can look like and what can be claimed, right?

Then we will go through the process to get labels created, get those on-products package, et cetera. That takes a period of time. We would then go into early experience program. And the reason to do that is although pain is an established market from a dog perspective, having a monoclonal antibody and

addressing nerve growth factor is still a new mechanism of action here. And getting the product in the hands of KOLs, [INAUDIBLE] leaders, that starts to write on it and get comfortable with the product. And that tends to help get the products seeded into a lot more clinics a lot earlier, which will benefit the long term. So we intend to run an early experience program for those long-term benefits that we see. And so that will run before we do a full launch, which we've been saying so late in 2023. Again, these are some of the factors that again, haven't changed, but why we did not include Librela in the guidance role as these milestones are crossed.

SPEAKER 2: Anything you can point to either using your experience with cytopoint or using your experience with Librela OUS in terms of what penetration we can expect in year one, year two, year three. It's 5% market penetration and first full year reasonable? It could be higher than that.

WETTENY JOSEPH: Look, I won't quote a number, but if you look at our patient share gains on Librela across Europe, those continue to climb and have really been above our expectations. I think in general, the US tends to ramp up a bit faster than Europe. So seeing what we saw in Europe gives us encouragement in terms of what may happen here in the US.

And so we're very optimistic in terms of what we'll see once the product launches. And again, we're going to be disciplined and deliberate about the steps that we take because we've learned from these prior launches whether you go back to cytopoint or if you look at more recently, with Librela across Europe, in terms of what's the play that works best again to get the product in the hands of care wells, and across as many clinics.

Then once the product is, then we can certainly start to do DTC on them, which drives even more patients into the clinic for this innovative solution.

SPEAKER 2: Anything you can say in terms of margin profile for monoclonal antibody? Apples to apples versus a small molecule when it's at scale?

WETTENY JOSEPH: Look, I think at scale, you're looking at MABs being accretive to the overall margin profile for the company. As with any other product, when you first launch, you are having to ramp up capacity that you've built. You may see some more variances for manufacturing perspective and all those type of things.

This is typical as you know, from my prior life, I've seen this play out thousands of times. And so that tends to happen. So I think margins might need some time to settle out after the first year or so call it. And then by the time you get fully into late second year, third year, you start to see margins that are at levels that are above the company average. But initially, they may be slightly dilutive given what I just described. But MABs in general, will be above the company's average when you set it out.

SPEAKER 2: OK. That's helpful. When you talk about being accretive after a year two or three, is that on a global basis? Because if you're using OUS facilities to manufacture Librela. You've already been in Europe for a year and a half, so are you already-- point being, once you get approved in the United States, it doesn't reset the clock, right? You're still at the point where you're ramping further?

WETTENY JOSEPH: Look, I won't get into that much detail on this specific product. I would use the same general rule of thumb that I just gave you. In general, there are other factors involved in terms of expansions and so on, and additional facilities that are involved so I won't get into too much more than that. But I would use the same rule of thumb here

SPEAKER 2: OK. I had to try it. All right. So let's talk about [INAUDIBLE] I don't know if you've heard, there might be competition coming. So-- yeah. Surprising, I know. I think you can say in terms of your

latest assumptions for timing of that approval and launch, and also now that at [? MVMX, ?] we've seen what the active ingredients are, how does that impact your thinking about what happens when that competitor hits the market?

WETTENY JOSEPH: Yeah. Look, we've been expecting competition for a TRIO for some time, right? We're now into our third, if not, we're going to be crossing into our fourth year on the product that we've been in the market, we're first to market in the largest parasiticides market in the world, in the US. That's a benefit that will stay with us even after competition. We expect, for the parasiticides market, which is about 6 billion globally. Call it 4 billion or so US and the rest outside us.

We expect the market to continue to expand. And this is still a relatively new standard of care where you have triple combination. And having heartworm incorporated should drive compliance as well, which is prevalent across the US. And so we expect-- this trend has been going on for probably eight years now, going back to when oils were introduced to continue to shift away from collars and topicals into oils, and now into triple combinations.

That is still in the early innings, if you will, and we expect that to continue to drive tailwind for this industry and this segment for some time to come. And the fact that we're first to market is also important. One thing I would say-- look, we'll see what the label is when the competition comes. I will stress, though, that TRIO, given heartworm is among the parasiticides that it covers, heartworm is the deadliest.

And we have 100% efficacy on heartworm after the first dose. That's an important factor that I want to stress with respect to our products. And again, we'll see what the competition comes out with, but we're very confident in our position and our ability to continue to grow TRIO.

I did cover earlier the fact that our supply recovery in 2022 may drive some variability throughout the year, including in the first quarter given the fact that both supply recovery and price rebuys caused inventories to be a little bit more elevated coming out of 2022 into 23. We will see that play out in the first quarter, to put some pressure on what the TRIO numbers are in the first quarter. That's already factored into our guidance and factored into what I said.

The other thing that we said on the call a couple of weeks ago is, we will expect some relatively aggressive promotional activity from competition. In fact, we're already seeing some that are ahead of their product launch in the first quarter. That's to be expected. We don't expect the competitive dynamic here, long term to be on price.

For reasons we've already stated, the fact that our trial is priced favorably versus the other two products to cover flea, tick and heartworm. So we believe we're well positioned there. We're first to market, which has first-to-market advantages. And we've been in the market for three years, with very high satisfaction levels across our customer base as well.

So again, I won't cover more on this but we expect to continue to grow. TRIO will be a significant contributor to growth for us in 2023, but we will see some variability, quarter to quarter, for the reasons I've stated, including somewhat aggressive promotional activities. But we're going to remain disciplined in terms of our response given our confidence in our product and everything else I've said.

SPEAKER 2: Got it. Yeah. And you talked about significant contributor to growth in 23. I think just given the growing base of the product, not as much of an incremental growth contributor as it was in prior years just off the bigger base. You have three or four competitions harder but still well above company average growth, right?

WETTENY JOSEPH: Yeah. Exactly. So I would-- TRIO grew 58% in 2022. OK? Even without competition, we wouldn't expect it to continue to grow at that pace. But I would expect significant contribution, which by definition, says it would be above the 6% to 8% range that we're giving for the company on the year. I won't get into more specifics than that on this specific product, but again, you'll see some variability from quarter to quarter. Those are factored already into our thinking.

SPEAKER 2: Got it. Yep. That seems very reasonable. We got about 10 minutes left and there's still a lot of topics to cover, so I want to pivot a little bit, just staying on innovation and new product, but let's take it to the next products.

What are the main innovations that you think we don't have yet? You talked about some areas of unmet need. This year, you're guiding to a pretty significant step up in R&D. It's 20% year over year growth in R&D. So what's your game plan beyond TRIO and beyond Librela [INAUDIBLE].

WETTENY JOSEPH: Yeah. Look, we have the portfolio in animal health as it is. But clearly, innovation has been a key driver of value creation for Zoetis and remains extremely important to us. And so we prioritize R&D in terms of our capital allocation.

If you look at the growth rate for R&D spend in the prior year, it was roughly in line with our revenue growth. So call it 8% operational growth in expense on R&D. In 2023, as we said on the earnings call, you're going to see a growth rate in R&D expense. It's about 2 and 1/2 times the midpoint of our growth rate for the company. That puts it somewhere north of 7%. Call it 7 to 7 and 1/2 of our revenues.

Now, we have had years in the past where we hit 7 and 1/2 before. So if you go back to-- I think 2019 might have been around that range. Now, we're talking about a higher based on revenue standpoint, so it's a much bigger spin, of course. But we have hit that kind of territory before.

And the other thing that I would say is the way we approach R&D spend is in a front loaded, we're going to spend X. We let the pipeline dictate what the spend is, and we have a very disciplined approach looking at ROI, looking at probability of success from a technical perspective or [INAUDIBLE] perspective. And we have stage gates that we run things through.

So when you see us have a meaningful increase in our R&D spend, it isn't dictated upfront. It is based on what the pipeline is and what the movements are across. Now, in terms of unmet needs you ask, right? Look, there's been surveys done for vet practitioners to ask, what are the unmet needs? And I think you can imagine that as the leader in animal health, we're going to be investing in most, if not all of those areas.

So the areas that we're talking about, we're still innovating across OA pain [INAUDIBLE]. Diabetes is an area that I will put on the list. I'll put allergy in general, in addition to just dermatology. But allergy, broadly speaking. Chronic kidney disease, cardiac diseases. These are all areas that we have interest in.

And I think one more thing that I would say is this, starting about five years ago with cytopoint launching and seeing the ability and the comfort of general practitioners across vet practices to be comfortable with a monoclonal antibody injection, that did give us an opportunity to step back and look at, what are the other areas that we can use this type of technology to address unmet needs?

And that's driving a lot of why we say MABs as a platform for us, that we'll continue to innovate in across the board. And then if you look at livestock, you've got vector vaccines across poultry that we started to launch and we'll continue to do immunotherapies as an alternative to antibiotics on the livestock side are areas that we're investing in.

And so we've always had a fairly broad remit, but we have a very disciplined approach across addressing these unmet needs and a track record in R&D, in innovation, that I think, still stands for itself. So very, very excited about what's to come in these areas that are driving the need to spend at the levels that we are spending in 2023.

SPEAKER 2: OK. And then both on the R&D front and on the capex front, and you've got a capex of up to a billion dollars. That's also a very sizable step up year over year and as revenues. I'm not going to ask for 2024 numbers, we just got it for 2023, but just help us think about is this a one-year investment? Is this a three-year investment? Is this a 10-year?

Sort of longer term. I know you guys don't have a long-term margin targets either, but how do we think about this spend stretching over multiple years?

WETTENY JOSEPH: Look, the way we look at it, if you look at capex, we're guiding to somewhere between 950 and a billion. That's roughly a 50% increase to what we spend in 2022. And that's largely driven by growth capex. Again, the MAB platform that we spoke of being a key driver there across the [INAUDIBLE] solid base as well if you look at growth across existing products, [INAUDIBLE] and all those type of things.

So there are various areas that we are spending. These projects tend to be multi-year projects. So I wouldn't sit here and say this is a one-time, one-year phenomenon. I could see this being a two to three-year range of this level of spend. And this level of spend from a percentage of revenue perspective if you look at it that way.

I would see that start to come back to normalized levels after that, but the dollars may still be somewhat elevated just the percentages given growth rate on revenue would start to come down. So hard to give you any more precision than that Mike, but I would not characterize this as a one-year event.

SPEAKER 2: OK. That's helpful. I'm going to try to go with rapid fire in the last couple of minutes to squeeze a couple more in.

WETTENY JOSEPH: Yeah.

SPEAKER 2: Can you say quickly about the diagnostics business? How that's faring versus competition? You have the sales force restructure, where are you in that process?

WETTENY JOSEPH: Yeah. Look, we made the change in the middle of last year after the first quarter. We continue to see really strong growth in international, growing above market. Clearly, this change has been disruptive to the US. We saw declines in the US in the second and third quarter. The fourth quarter also declined but better than the second and third quarter, relatively speaking.

I think we'll continue to see some incremental improvement, but still down in the first half of 23 with growth in the second half of 23 from a US perspective, but continue to see solid growth in international throughout as we have seen last year and the years before.

SPEAKER 2: And then longer term, is growth with the industry for DX a fair starting point for [INAUDIBLE]?

WETTENY JOSEPH: Yeah. I think that's a fair starting point. We have demonstrated an ability to grow above market outside the US. I think the US will position it at market, which we've done in the past. But certainly, with this investment, we would expect to have some incremental benefit coming out of that given we have innovation across not only our X, but DX that we continue to work on across images and other areas that we believe will continue to propel us.

At least at a market if not better than market, long term, which again, DX rose faster than the broad animal health market to begin with.

SPEAKER 2: Got it. And maybe one last one on derm really quickly. There's a lot of noise throughout 2022 on how derm was performing. I mean some of that fluctuation quarter to quarter. I think for the year, you ended at 17% operational, so. Expectations going forward? Again, bigger base, but both on the competition front and also on the derm franchise as it is.

WETTENY JOSEPH: Yeah. Look, derm, we just continue to see really strong growth even as [INAUDIBLE] is approaching 10 years inside of 0.5. These are products that continue to have opportunity to grow with \$6 dogs in the US, that we estimate that are untreated and another 2 million that are under-treated. So we continue to see room. And internationally, there was 17% operational growth for the year, international grew 27% on the year and we continue to see substantial more room to drive growth across derm.

Look, we grew 17% operationally, after a 23% growth year and the [INAUDIBLE] growth over the prior two years. I wouldn't expect that level of growth, but I would expect double-digit growth to continue to come from our derm franchise. And we're not expecting competition to be launched in 2023 across derm. So another year of continuing to drive. And even beyond competition, we expect to continue to grow the product. We continue to innovate across this franchise as well.

SPEAKER 2: OK. Well, I think I've been asking about competition in derm for about five years, so it sounds like I'll get to ask it again next year, so stay tuned. Wetteny, plenty more we can touch on but unfortunately, we're out of time. So we're going to have to call it there. Thanks so much for joining us. Always a pleasure to chat and catch up.

WETTENY JOSEPH: Thanks for having me, Mike. Been my pleasure. Thank you. Enjoy the rest of the conference.

[MUSIC PLAYING]